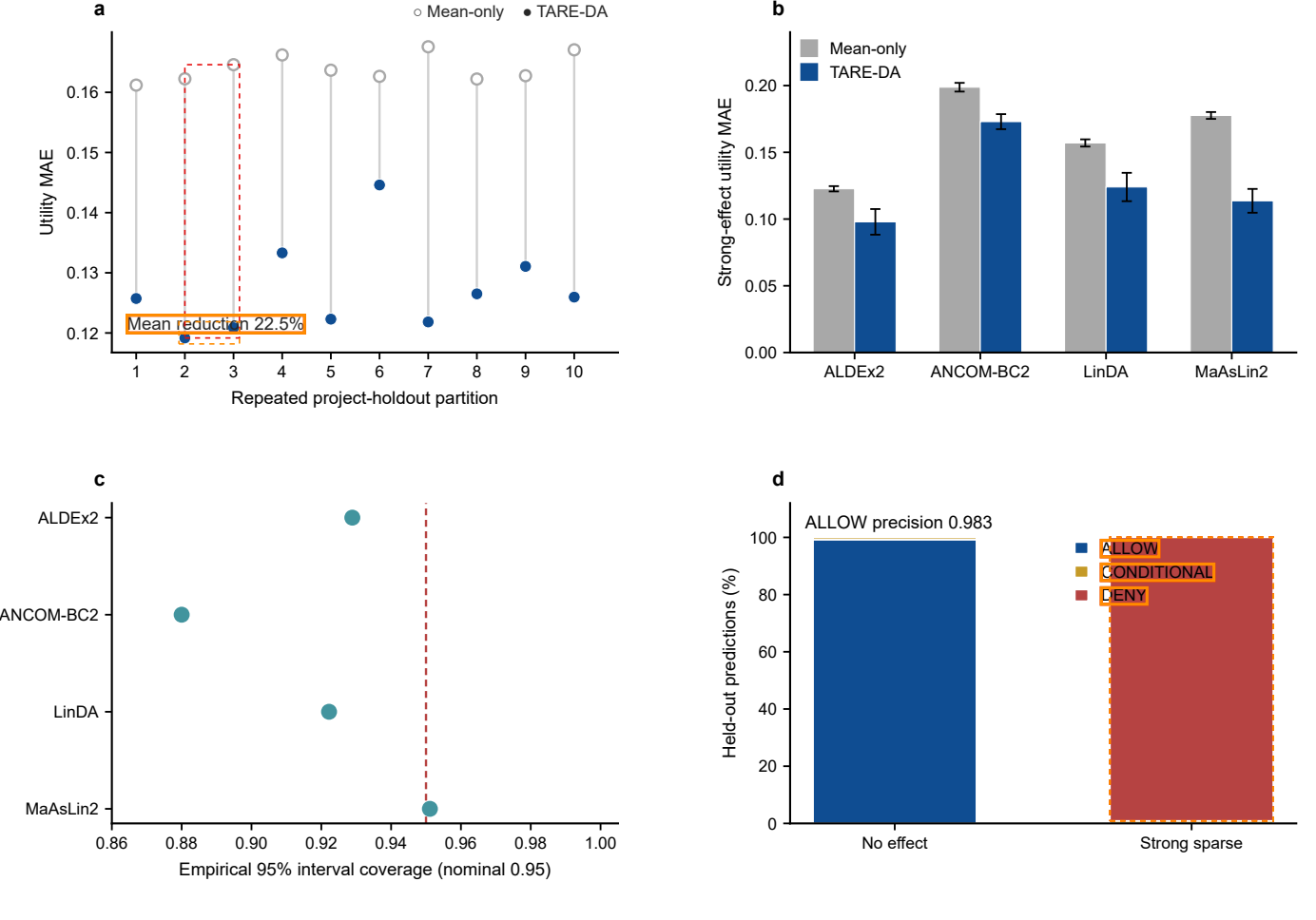

a
Mean-only
TARE-DA
Utility MAE
0.16
0.15
0.14
0.13
0.12
Mean reduction 22.5%
1
2
3
4
5
6
7
8
9
10
Repeated project-holdout partition
b
Mean-only
TARE-DA
Strong-effect utility MAE
0.20
0.15
0.10
0.05
0.00
ALDEx2
ANCOM-BC2
LinDA
MaAsLin2
c
ALDEx2
ANCOM-BC2
LinDA
MaAsLin2
0.86
0.88
0.90
0.92
0.94
0.96
0.98
1.00
Empirical 95% interval coverage (nominal 0.95)
d
ALLOW precision 0.983
ALLOW
CONDITIONAL
DENY
Held-out predictions (%)
100
80
60
40
20
0
No effect
Strong sparse